Light

GCSE Computer Science – Programming Fundamentals (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the character data type used for?

- ○ **A.** Storing a single letter, digit, or symbol
- ○ **B.** Storing multiple words of text
- ○ **C.** Storing decimal numbers
- ○ **D.** Storing True or False values

Q2 (1 mark)

What is a syntax error?

- ○ **A.** An error where the program runs but produces incorrect results

- ○ **B.** An error where the program runs too slowly
- ○ **C.** An error caused by incorrect user input
- ○ **D.** An error where the code violates the grammatical rules of the programming language, preventing the program from running

Q3 (1 mark)

What is a logic error?

- ○ **A.** An error that prevents the program from starting
- ○ **B.** An error in the code's spelling or punctuation
- ○ **C.** An error where the program runs but produces incorrect or unexpected results due to flawed reasoning in the algorithm
- ○ **D.** An error displayed by the operating system

Q4 (1 mark)

A program accepts ages between 18 and 65 inclusive. Which values would be classed as boundary test data?

- ○ **A.** 20, 40, 50
- ○ **B.** 0, 100, -5
- ○ **C.** 18 only
- ○ **D.** 17, 18, 65, 66

Q5 (1 mark)

What is normal (typical) test data?

- ○ **A.** Data that is clearly within the acceptable range and represents typical expected input
- ○ **B.** Data at the boundaries of the acceptable range
- ○ **C.** Data that is outside the acceptable range
- ○ **D.** Data that causes the program to crash

Q6 (1 mark)

Which type of iteration is a FOR loop?

- ○ **A.** Condition-controlled iteration
- ○ **B.** Infinite iteration
- ○ **C.** Random iteration
- ○ **D.** Definite (count-controlled) iteration

Q7 (1 mark)

Which type of iteration is a WHILE loop?

- ○ **A.** Indefinite (condition-controlled) iteration
- ○ **B.** Definite (count-controlled) iteration
- ○ **C.** Sequential iteration

- ○ **D.** Parallel iteration

Q8 (1 mark)

Which relational operator checks if two values are NOT equal?

- ○ **A.** ==
- ○ **B.** =
- ○ **C.** >=
- ○ **D.** !=

Q9 (1 mark)

What does the following statement do? `score <- 100`

- ○ **A.** Assigns the value 100 to the variable called score
- ○ **B.** Checks if score equals 100
- ○ **C.** Outputs the value 100
- ○ **D.** Creates a constant called score

Q10 (1 mark)

In most programming languages, what is the position (index) of the first character in a string?

- ○ **A.** 1

- B. 0
- C. -1
- D. The length of the string

Q11 (1 mark)

What is the purpose of an INPUT statement in programming?

- A. To receive data from the user, typically via keyboard
- B. To display information on the screen
- C. To perform a calculation
- D. To end the program

Q12 (1 mark)

What is the difference between a procedure and a function?

- A. Procedures are faster than functions
- B. Functions can take parameters but procedures cannot
- C. Procedures can only be used once in a program
- D. A function returns a value to the calling code, while a procedure performs actions but does not return a value

Q13 (2 marks)

What is output when `score` is 75?

```
IF score >= 90 THEN
    OUTPUT "A"
ELSE IF score >= 70 THEN
    OUTPUT "B"
ELSE IF score >= 50 THEN
    OUTPUT "C"
ELSE
    OUTPUT "F"
ENDIF
```

- ○ **A.** A
- ○ **B.** B
- ○ **C.** C
- ○ **D.** F

Q14 (2 marks)

A program needs to convert a total number of minutes into hours and remaining minutes. If `totalMinutes = 135`, what are the values of hours and minutes after this code?

```
hours <- totalMinutes DIV 60
minutes <- totalMinutes MOD 60
```

- ○ **A.** hours = 2, minutes = 15
- ○ **B.** hours = 2.25, minutes = 0
- ○ **C.** hours = 2, minutes = 35
- ○ **D.** hours = 135, minutes = 60

Q15 (2 marks)

Given `text <- "COMPUTING"`, what is the result of extracting a substring starting at position 0 with length 4?

- ○ **A.** MPUT
- ○ **B.** OMPU
- ○ **C.** COMP
- ○ **D.** PUTI

Q16 (2 marks)

What is the value of `count` when this loop finishes?

```
count <- 0
REPEAT
    count <- count + 2
UNTIL count >= 10
```

- ○ **A.** 8
- ○ **B.** 12
- ○ **C.** 10
- ○ **D.** 2

Q17 (2 marks)

For the condition `(age >= 18) AND (hasTicket = True)` to evaluate to True, which of the following must be correct?

- ○ **A.** Either age >= 18 OR hasTicket = True
- ○ **B.** Both age >= 18 AND hasTicket = True must be satisfied
- ○ **C.** Only age >= 18 needs to be true
- ○ **D.** Only hasTicket = True needs to be true

Q18 (2 marks)

What are parameters used for in subroutines?

- ○ **A.** To pass data into a subroutine so it can work with different values each time it is called
- ○ **B.** To make the subroutine run faster
- ○ **C.** To store the result of the subroutine permanently
- ○ **D.** To count how many times the subroutine has been called

Q19 (2 marks)

Why might you need to convert a string to an integer in a program?

- ○ **A.** To make the string display in uppercase
- ○ **B.** Because input from a user is often received as a string, but arithmetic operations require numeric data types
- ○ **C.** To store the string in less memory
- ○ **D.** To make the program run faster

Q20 (2 marks)

An array called `colours` contains `["Red", "Green", "Blue", "Yellow"]`. What value is stored at `colours[2]`?

- ○ **A.** Red
- ○ **B.** Green
- ○ **C.** Yellow
- ○ **D.** Blue

Q21 (2 marks)

Why is it important to use meaningful identifier names in programs?

- ○ **A.** Meaningful names make the program run faster
- ○ **B.** The programming language requires names to be at least 10 characters long
- ○ **C.** Meaningful names make code easier to read, understand, and maintain for both the original programmer and others
- ○ **D.** Meaningful names reduce the file size of the program

Q22 (2 marks)

A form requires a UK postcode. Which of the following is an example of erroneous test data?

- ○ **A.** SW1A 1AA
- ○ **B.** M1 1AE

- ○ **C.** B33 8TH
- ○ **D.** 12345

Q23 (2 marks)

What is the primary use of random number generation in programming?

- ○ **A.** To make programs run at unpredictable speeds
- ○ **B.** To introduce variability and unpredictability, such as in games, simulations, or selecting random items
- ○ **C.** To encrypt all data in the program
- ○ **D.** To automatically fix errors in code

Q24 (3 marks)

What are the final contents of the array **grid** after this code executes?

```
grid <- [[0, 0], [0, 0]]
FOR row <- 0 TO 1
    FOR col <- 0 TO 1
        grid[row][col] <- row + col
    ENDFOR
ENDFOR
```

- ○ **A.** [[0, 0], [0, 0]]
- ○ **B.** [[0, 1], [1, 2]]
- ○ **C.** [[1, 2], [2, 3]]

- D. [[0, 0], [1, 1]]

Q25 (3 marks)

What is output by the following code?

```
FUNCTION triple(n)
    RETURN n * 3
ENDFUNCTION

FUNCTION addTen(n)
    RETURN n + 10
ENDFUNCTION

a <- triple(2)
b <- addTen(a)
c <- triple(b)
OUTPUT c
```

- A. 26
- B. 36
- C. 48
- D. 66

Q26 (3 marks)

A subroutine validates that a password is at least 8 characters long and contains at least one digit. What is the output when called with `validatePassword("secure99")`?

```
FUNCTION validatePassword(password)
    IF LENGTH(password) < 8 THEN
        RETURN "Too short"
    ENDIF
```

```
    hasDigit <- False
    FOR i <- 0 TO LENGTH(password) - 1
        IF password[i] >= "0" AND password[i] <= "9" THEN
            hasDigit <- True
        ENDIF
    ENDFOR
    IF hasDigit = False THEN
        RETURN "No digit"
    ENDIF
    RETURN "Valid"
ENDFUNCTION
```

- **A.** Too short
- **B.** No digit
- **C.** Valid
- **D.** False

Q27 (3 marks)

A school stores student data using records with fields: `name` (String), `age` (Integer), and `yearGroup` (Integer). Which declaration correctly defines this record structure?

- **A.** student <- [name, age, yearGroup]
- **B.** RECORD Student name: String age: Integer yearGroup: Integer ENDRECORD
- **C.** Student = name + age + yearGroup
- **D.** ARRAY Student[name, age, yearGroup]

Q28 (3 marks)

The ASCII code for 'A' is 65, and letters are sequential (B=66, C=67, etc.). What does this code output?

```
letter <- "D"
code <- ASC(letter)
newCode <- code + 3
newLetter <- CHR(newCode)
OUTPUT newLetter
```

- ○ **A.** D
- ○ **B.** G
- ○ **C.** 68
- ○ **D.** 71

Q29 (3 marks)

A theme park ride has these requirements: riders must be at least 140cm tall, aged 8 or over, and must not have a heart condition. Given: `height = 150`, `age = 10`, `hasHeartCondition = True`, what does this code output?

```
IF height >= 140 AND age >= 8 AND NOT hasHeartCondition THEN
    OUTPUT "Allowed"
ELSE
    OUTPUT "Not allowed"
ENDIF
```

- ○ **A.** Allowed
- ○ **B.** Not allowed
- ○ **C.** True

○ **D.** Error

Q30 (3 marks)

A programmer divides a large project into modules: `getUserInput()`, `processData()`, `saveResults()`, and `displayOutput()`. Another programmer needs to modify how results are saved. Analyse how the modular structure benefits this maintenance task.

○ **A.** The programmer must still read and understand the entire codebase before making any changes

○ **B.** The modular structure provides no benefit because all modules are interconnected

○ **C.** The programmer can focus on the saveResults() module without needing to understand or modify the other modules, reducing complexity and risk of introducing bugs elsewhere

○ **D.** Modular programming makes maintenance slower because of the overhead of multiple files

Total: **55** marks • Time: **30 min**

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